Table 21.11 Characteristics of some solidified high level waste products

Property	Fluidized bed calcine	Phosphate glass	Borosilicate glass
Physical form	Granular, 0.3–0.7 mm	Monolithic	Monolithic
Bulk density (kg m^{-3})	1.0–1.7	2.7–3.0	3.0–3.5
Maximum weight % FP	50	35	30
Thermal conductivity (W m^{-1} K^{-1})	0.2–0.4	1.0	1.2
Leachability (20°C, kg m^{-2} d^{-1})	1–10	10^{-5}–10^{-3}†	10^{-6}-10^{-4}‡
Maximum center temperature (°C)	550††	400*	700*
Maximum allowable heat density (kW m^{-3})	45 (#)	53	127
	38 (@)	36	92

†Devitrified (crystalline) glass has leach rates 10^{-2}–10^{-1} kg m^{-2} d^{-1}.
‡At 100°C the leach rate is ~10^{-2} kg m^{-2} d^{-1}.
††Because of risk for FP volatilization.
* Devitrifies at higher temperature.
#Forced water cooling of a cylinder; Ø 0.3 m, surface temperature ≤100°C.
@Natural air cooling of a cylinder; Ø 0.3 m.

FPs) in a single cylinder. The high heat conductivity makes it feasible to use short fuel cooling times (down to 0.5 y) and diminishes the demand for interim fuel element storage basins or HLLW storage tanks. The solidified waste glass is collected in stainless steel cylinders.

A typical waste canister may contain ~9% FPs, emit ~10 kW after 1 y, ~1 kW after 10 y, and ~0.5 kW after 40 y. They can be stored in air-cooled vaults or water-cooled pools. Before final deposition the HAW containers are reconditioned, i.e. enclosed in an additional canister of type described below. At the higher power level they are stored in water-filled pools. After >10 y air-cooled vaults may be satisfactory; forced cooling is usually considered, but with precaution for sufficient convection cooling in case of ventilation failure.

21.13. DEPOSITION IN GEOLOGIC FORMATIONS

A general consensus has developed that disposal of radioactive waste should be in the country in which the nuclear energy is produced. Thus, even if the spent fuel elements are sent from one state to another for reprocessing (like spent fuel from Japanese reactors being sent to the Sellafield reprocessing plant in the United Kingdom), the producer state (Japan) must guarantee its readiness to accept the processed waste. This condition has led to intensive national investigations of the safest way to dispose of the high level waste, either in the form of solidified HLLW, or spent fuel elements in the once-through option. In these studies geologic repositories have been the prime consideration, e.g. crystalline silicate rock, clay layers, salt domes, the sea bed, etc.

21.13.1. Properties of geologic formations

Many geologic formations are being studied for final storage of the waste canisters: rock salt, crystalline rock (e.g. granite), volcanic tuff, clay, etc. The main requirements are:

(i) geologic stability, i.e. regions of very low seismic, volcanic, or other geologic activity;
(ii) absence of large fractures, holes, etc.;
(iii) impermeability to surface water;
(iv) negligible groundwater circulation with no flow-lines leading to nearby potential intake sources;
(v) good heat conductivity;
(vi) of little interest to and as remote as possible from human activities.

There are large geologic formations, which are very old (many hundred million years) and geologically stable, e.g. the Scandinavian and Canadian shields. These are usually fractured, even if considerably large volumes can be found (10^6 m^3) which are free of fractures. A nuclear waste storage facility will, however, for reasons given below, cover rather large areas, several km^2; thus for crystalline rock fracture zones, which carry ground water, must be taken into account. Rock salt and clay formations, on the contrary, are usually free of fractures: rock salts are dry, but clay formations are usually wet, though the water migration through the formation is extremely small (see below).

A limiting factor for geologic disposal is heat conductivity. If the thermal conductivity of the geological deposit is too poor, the material in the waste canisters may melt and react with the encapsulation, possibly destroying it. Because a safe disposal concept must rely on conduction cooling, the relatively low heat conductivity of the geologic deposit makes it necessary to disperse the waste material over a large volume. The thermal conductivity of $\sim$3 W m^{-1} K^{-1} for granitic rock (compare rock salt $\sim$9 and wet clay $\sim$1) limits the waste heat density of the rock wall to $\leq$20 W m^{-2}. Therefore storage repositories would consist of long tunnels or deep holes at large interspace. Three concepts are under investigation: (i) very deep holes (VDH, up to 3000 m) in which waste canisters would be stacked, (ii) deep repositories (300–500 m) with parallel horizontal tunnels of moderate length ($\leq$100 m), 20–100 m apart, upper part in Figure 21.25, and (iii) deep repositories with a few very long tunnels (>1 km, VLH).

With a typical heat production of $\sim$500 W per canister containing vitrified HAW (at 40 years cooling) the wall heat density would be 6–7 W m^{-2}. The internal temperature of a glass canister would then never exceed 90°C, and the surface temperature (above the storage holes, Fig. 21.25) not more than 65°C. Slightly higher temperatures can be accepted for spent fuel elements.

Such deep (geologically) repositories are safe against surface activities (including nuclear explosions) except drilling or mining. In order to avoid such an occurrence after the repository has been forgotten, it should be located in a formation of no interest to

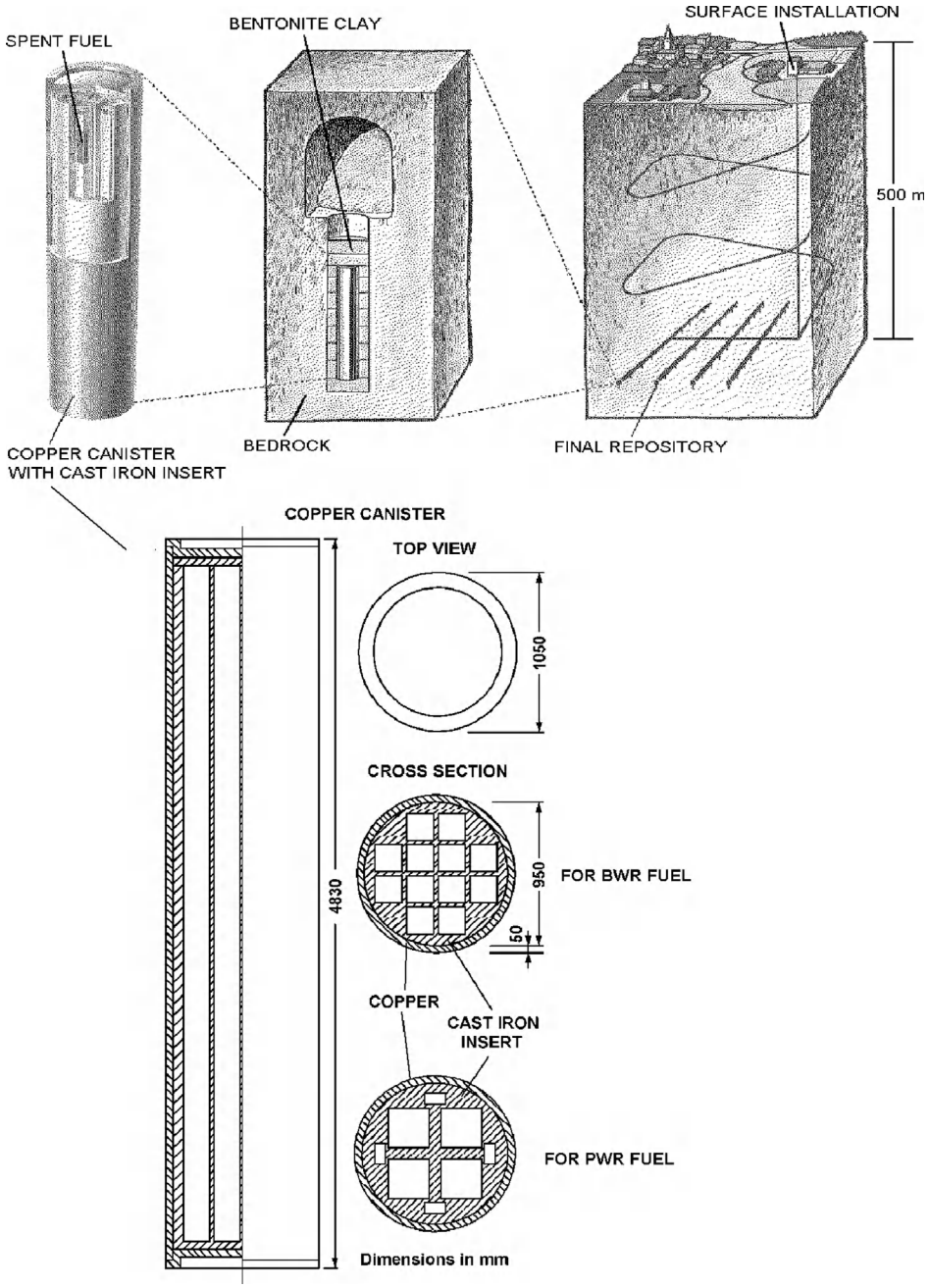


Figure 21.25 Principles of storage of HLW: upper part, Swedish concept; lower part, canister with inserts for final storage of spent BWR fuel and spent PWR fuel. *(From SKB.)*

society; e.g., the formation should not contain any valuable minerals. From this point of view, location in deep seabeds has been suggested. Such a location can be chosen either deep in the bottom silt or in a bedrock under the sea floor, or, possibly, in a tectonically active trench, where with time the waste would be pulled down into the interior of the earth.

The risk for a land-based geologic repository is that a combination of more or less unpredictable circumstances could lead to rupture of the protective barriers around the waste matrix, followed by dissolution and transport of the most hazardous products by water to a place where it can enter into the food chain. Therefore a clay layer (e.g. bentonite) between the canister and the rock wall is suggested, partly to act as a mechanical buffer, so that even considerable slippage caused by earthquakes would have little effect on the mechanical integrity of the canister and partly to reduce water flow around the canister.

Most rocksalt deposits are very old. In many cases they rise in the form of a dome, the top of which may be at a few hundred meters below surface level, and with a depth up to more than 1000 m. The dome is usually protected from groundwater by a calcite ($CaCO_3$) cap. Salt has been mined in such formations for centuries. The formation contains only microcrystalline water and is extremely dry (otherwise it would have dissolved during the geologic ages). Therefore, canisters emplaced in such formations are not believed to dissolve (as long as no water enters through the hole mined through the calcite layer). The good heat conductance allows emplacement of waste of higher power density than for deposits in clay or granite. Since the salt is plastic, the holes and corridors, which are backfilled with crushed salt, selfseal and no clay buffer is needed. The canisters would probably be crushed in the plastic salt, and therefore could be irretrievable after a hundred years.

Granite and clay formations are percolated by groundwater except at depths which presently are considered impractical. The water flow rate is given by *Darcy's law*

$$F_w = k_p i\, S \left(\mathrm{m^3 s^{-1}}\right) \tag{21.15}$$

where k_p is the permeability (m s^{-1}), i the hydrostatic gradient (m m^{-1}), and S the flow cross section (m^2). A typical value for Scandinavian granite formations at 500 m depth is $F_w = 2 \times 10^{-4}$ m^3 m^{-2} y^{-1}, which gives a ground-water velocity of 0.1 m y^{-1}, and a time of 5000 y for the water to move 500 m. In this case the rock permeability is taken as 10^{-9} m s^{-1}; rock formations closer to the surface usually have higher permeabilities, but many formations with much lower values are also known ($\sim 10^{-13}$).

Clays consist of small particles, usually <2 μm and with an average size of <0.1 μm, of various minerals like quartz, feldspar, montmorillonite (a hydrated aluminum silicate with high ion-exchange capacity), mica, etc. The overall chemical composition is mainly a mixture of silica, alumina, and water. The small particles in the clay give rise to a very large surface area (1 cm^3 of particles of 0.1 μm diameter have a total surface area or about

60 m^2) and correspondingly high sorption capacity for (eventually) dissolved waste products. Although the clay can take up large amounts of water (up to 70%) without losing its plasticity, the water permeability is extremely low. For the sodium bentonite clay (~90% montmorillonite, plenty available) considered in many projects, the permeability is 2×10^{-14} m s^{-1} (10% water in clay compacted to a density of 2100 kg m^{-3}). Such clay can be considered impermeable to groundwater. Natural clays of this type occur in many countries (the Netherlands, Italy, etc.).

21.13.2. Waste conditioning before final storage

The basic philosophy in storing high levels of radioactivity in geologic formations is the use of a multi barrier system to protect it against dissolution. These barriers are:

(i) The waste matrix itself, which is made highly insoluble; this is achieved by vitrification of the reprocessing HLLW as described above, while the spent unreprocessed fuel elements keep the radioactive products in a highly insoluble UO_2 matrix.

(ii) The waste matrix (glass or UO_2) is encapsulated in an "insoluble" canister.

(iii) The canister is surrounded by a clay buffer in the case of a crystalline rock deposit.

Here we shall only discuss the encapsulation, for which numerous concepts have been suggested.

(a) *Vitrified waste.* The stainless steel cylinders containing the vitrified radioactive waste are not considered to provide sufficient protection against long term corrosion in the final repository. Additional encapsulation in various corrosion resistant materials is therefore suggested: lead, titanium, copper, gold, graphite, ceramics, etc. For example, in one proposal canisters containing 170 l vitrified waste in a 3 mm thick stainless steel cylinder will be surrounded by 100 mm lead in a casing of 6 mm titanium; the overall dimensions of the cylinder will be 1.6 m long with a diameter 0.6 m. One such cylinder contains solidified HLLW from reprocessing 1 t spent fuel from a 1000 MW_e light water reactor; this corresponds to about 20 cylinders per year.

(b) *Spent fuel.* Although UO_2 dissolves more slowly than glass in groundwater, spent fuel elements must be recanned before entering the final storage facility. For this purpose single fuel elements or bundles are placed in cylindrical canisters. The lower part of Figure 21.25 shows an example. The outer part of the canister may consist of iron, stainless steel, copper, etc. selected to resist the repository environment (e.g. dry or wet). For storage in very long or deep dry holes, rounded caps are likely to be used. Waste canisters for unchopped spent fuel elements will be rather long, but it has also been suggested that the fuel is chopped as in the German Pollux concept, Fig. 21.26, which is designed for multi-purpose use, both to fit all kinds of spent fuel elements (from BWRs, PWRs, HTGRs etc) as well as vitrified high level waste. It is a double-shell concept, consisting of leak-tight welded steel; between the two steel

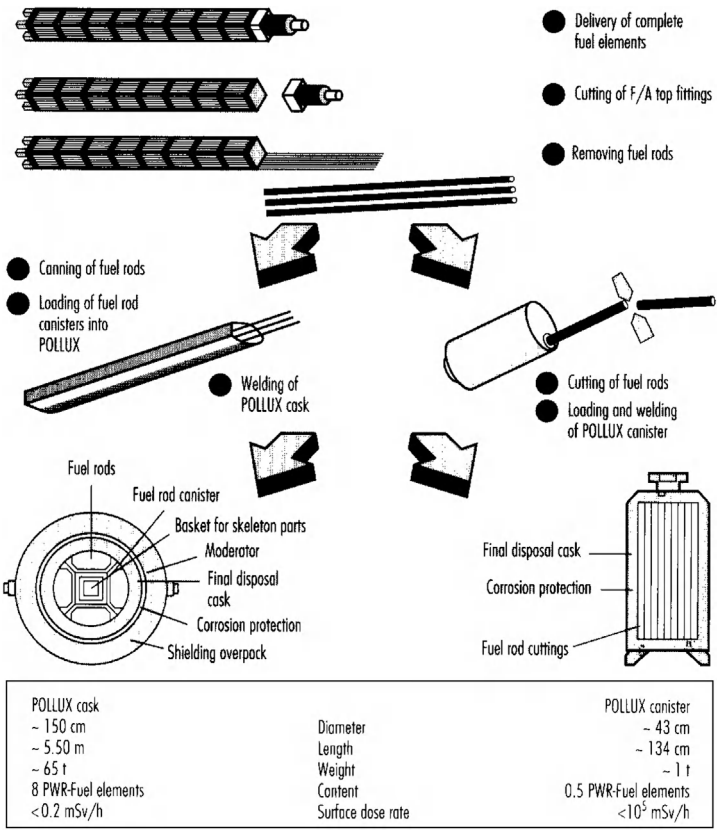


Figure 21.26 The proposed German POLLUX cask for final disposal of spent fuel and/or vitrified high level waste in salt.

walls is a corrosion resistent sheet of Hastelloy C4. The void is either gas filled (He) or filled with some suitable material like lead, which has good heat conductivity and also provides some radiation protection. Figure 21.25, shows the design of a spent fuel element canister where the fuel elements are inserted in a cast iron cylinder which is surrounded by 50 mm of copper metal. The thick metal layer has several purposes: (i) to simplify handling during deposition operations; (ii) to protect its content from mechanical deformation; (iii) to reduce the surface radiation to so low values that the radiolysis of groundwater will not contribute to canister corrosion; (iv) to protect, as long as possible, the waste-containing matrix from corrosion. For example, by using an outer copper wall, the lifetime of the canister is expected to greatly exceed 1000 y for oxygen containing groundwater, and 10^4 y for reducing groundwater (the lifetime of a 5 cm thick copper layer is now estimated to be $>10^6$ y). The dissolution of waste canisters and leaching of waste is discussed in §22.10.

21.13.3. Repository projects

In one concept the canisters are stored in dry areas (e.g. a desert floor) after having been surrounded by concrete. A space may be left between the inner cylinder and the surrounding concrete wall to allow air convection cooling. As long as the climate stays dry the pillars will erode very slowly and last for tens of thousands of years. This procedure also allows for easy retrievability. An alternative encapsulation is achieved by using hot ($\sim$1500°C) isostatic (100–300 MPa) compression to surround the fuel bundles by a homogeneous, dense ceramic material, like corundum (microcrystalline Al_2O_3) or graphite. Since corundum and graphite are natural minerals, the long term resistance should be very high, even against water.

The main concept is to store the vitrified HLW and the spent fuel elements irretrievably in deep underground geologic formations, according to designs described above. Many such formations are now being evaluated: rock salt (mainly Germany, the Netherlands and USA), crystalline rock (Canada, Finland, France, Japan, Sweden, USA), volcanic tuff (USA), clay (Belgium, Italy), etc. Cross sections of geologic repositories for HLW are shown in Figures 21.17(C) and 21.25. Also disposal into polar ice sheets (USA) and the seabed (UK, USA) have been investigated, but are no longer considered feasible.

The most advanced projects are (i) at the abandoned Asse salt mine in Germany, where drums with low and intermediate level activities have been deposited at 300 m depth in 1967–78; the repository is now used for research; a final repository in salt at Gorleben has been planned, (ii) the waste isolation pilot plant (WIPP) at a bedded salt deposit in New Mexico at 600 m depth; the planned repository is currently examined by the authorities for full scale operation; it will primarily be used for α-bearing military waste; in Yucca Mountain extensive studies have been made for a repository 300 m deep in dry tuff for civilian and defense nuclear waste (total capacity $\sim$7000 t IHM) – however, according to current political decisions it will not be used as a repository, (iii) the Swedish KBS project for disposal in granite; the Stripa mine has been studied for more than a decade by an international team and a new underground laboratory has been built at Äspö; other hard rock laboratories are in operation in Canada, Finland and Switzerland, and (iv) a laboratory at 250 m depth in clay at Mol, Belgium, has been in operation since 1983, etc. These sites are all located in geologic formations, which have been unaltered for more than 100 million years. From the tectonic plate theory continual tectonic stability is expected for at least the next 10 million years. Although ice ages may alter surface conditions at northerly located sites, they are expected to have a negligible effect on repositories at great depth (>500 m). In 2001 the Finnish parliament gave its permission to begin construction of a KBS-like final repository for spent fuel in granite. It will be the first of its kind in the world.

21.14. BENEFICIAL UTILIZATION OF NUCLEAR WASTES

The amount of spent fuel generated during 2003 amounts to about 8000 t. It contains large and potentially valuable sources of metals and radioactive nuclides. The known and predicted cumulative amounts are indicated in Figure 21.11. Though today considered a liability it may in the future become a needed asset. Since the extraction and utilization of some of the fission products or actinides will probably not be economic after the waste has been vitrified and placed in permanent geologic storage, the nuclear fuel reprocessing scheme should therefore be designed for byproduct extraction. Presently the most interesting products in the waste are the platinum group metals (due to their metal values) and ^{90}Sr, ^{137}Cs, ^{85}Kr, ^{237}Np/^{238}Pu, and ^{241}Am (due to their radiation properties, cf. §§8.11, 18.5, etc).

The waste contains considerable amounts of Ru, Rh, and Pd, all metals in scarce abundance on earth. These elements are used as catalysts in the chemical industry, for catalytic exhaust cleaning in cars, and as corrosion resistant materials. The United States demand exceeds the domestic production by about a factor of 100. The United States would be independent of import from year 2000 if these elements were recovered from the yearly generated spent fuel. This is particularly true if technetium is recovered, since it can often replace platinum. Some of the recovered elements would be radioactive, but the activities would be small enough to make the elements easy to handle.

Beta radiation from ^{85}Kr on phosphors causes visible light. Radiokrypton light sources have widespread applications where reliable lights are required as, for example, at airports, railroads, hospitals, etc., or where sources of electricity could cause dangerous explosions, as in coal mines, natural gas plants, etc. However, fission product krypton contains only 4% of ^{85}Kr, which makes it unsuitable for high intensity lightning applications. ^{85}Kr must therefore be enriched about a factor of 10, which presently can be done e.g. by thermal diffusion.

By the year 2000, over 100 MW heat will be produced by radiostrontium. Strontium fuelled thermoelectric generators are used in several countries to power unmanned weather data acquisition systems, lighthouses, and other navigation aids, etc. Their reliability surpasses any other remote power source. The current thermoelectric generators have a thermal-to-electrical efficiency of about 5%, while recently developed thermal-to-mechanical systems show efficiencies of 25–30%. The use of such systems could be expanded vastly by increased recovery of suitable fission products and actinides. However, due to environmental concerns many of the generators deployed so far at sea or on shores are now being recovered and scrapped.

Food sterilization by radiation is potentially of global importance (§8.11). Though this presently is done by ^{60}Co or accelerator radiation, the advent of large quantities of radiocesium recovered from nuclear wastes may have a considerable positive impact on the economics and scale of food irradiation. An almost equally important use of ^{137}Cs